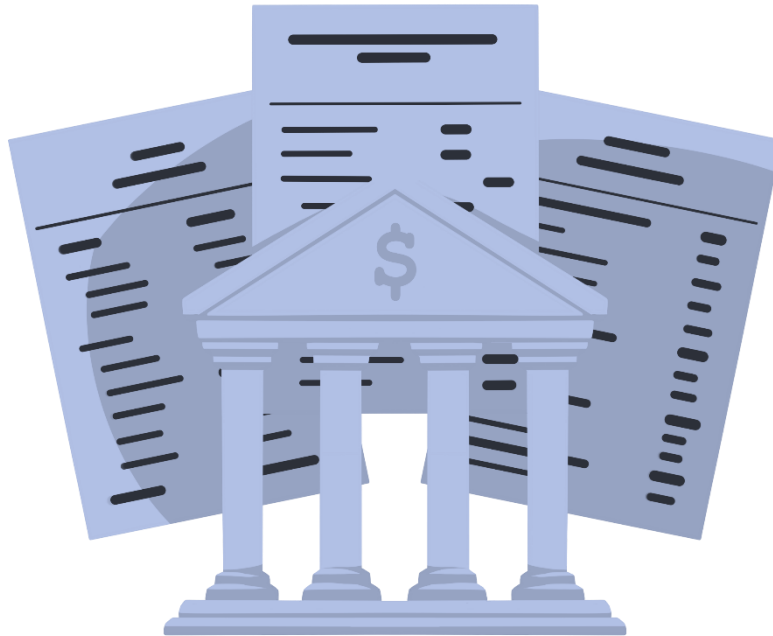


Accounting & Valuation Course



Lecture 2

Valuation Methods and Metrics

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Valuation Methods and Metrics

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Introduction

The Accounting & Valuation course aims to give you an overview of the financial techniques and resources used when it comes to valuing a company. In our previous lecture we build up the three financial statements, understanding what each component means, and where it comes from. The second section of this course will introduce the process of valuing a company and the key methodologies investors use to project future performance.

Comparison of the Three Financial Statements

Before we begin exploring the various methods of valuing a company, we will explore some key characteristics of the three financial statement and what purpose they serve to us as an analyst. The income statements give us an idea of the *income* (revenue minus expenses) a business has generated over a period of time (typically over a quarter or a year). It gives us an idea of the profitability of a company by using revenue, expenses, and gross profit to calculate net income. In contrast, the balance sheet shows us the level of assets, liabilities and shareholders' equity at a particular point in time. It gives us an idea of the financial position of a company by using the accounting equation ($\text{assets} = \text{liabilities} + \text{shareholders' equity}$) to demonstrate the levels of each component. Finally, the cashflow statement, like the income statement, is measured over a period of time (typically quarterly or annually) and outlines the overall cashflow into and out of a business. From the cashflow statement an analyst can get an idea as to how a business uses its cash.

The Concept of Valuation

The analytical process of determining the current (or projected) worth of an asset or company.

An analyst placing a value on a company looks at the business's management, composition of its capital structure (distribution of debt and equity in a company), the prospect of future earnings, and the market value of its current assets as well as other metrics. In this course we will introduce some common methods of valuation as well as some key definitions to help understand what valuing a company means and how it is conducted.

The first key idea to understand is fundamental analysis. This is the process of determining a stock's real or 'fair market' value. It aims to measure the securities intrinsic value based on the issuing company's financial situation and current market and economic condition. The concept of intrinsic value refers to the perceived value of a security based on future earnings or some other company statistics unrelated to the market price of a security (the price it is traded at). This is where valuation techniques come into play. Analysts conduct valuations to determine whether a company or asset is overvalued or undervalued by the market.

When it comes to valuations you tend to have two different types of models: absolute valuation models and relative valuation models.

Absolute Valuation Model

An absolute valuation model aims to find the intrinsic or 'true' value of an investment based only on the fundamentals. These models focus on factors such as dividends, cash flow, growth rate, for a **single** company without worrying about any metrics or performance of other companies. Models such as the dividend discount model (DDM) or discounted cash flow model (DCF) all fall into this category.

Relative Valuation Models

Relative valuation models compare the company of interest, to other companies typically of similar size and industry. These models usually involve calculating multiples and ratios (e.g., P/E ratio) before comparing them with multiples and ratios for other companies. This type of valuation tends to be much easier and quicker to calculate than absolute valuations, so they tend to be conducted first before a more complicated absolute valuation is conducted.

Valuation Methods

In this course we will cover the three most common types of valuation methodology: precedent transaction analysis, comparable company analysis, and discounted cash flow model.

Precedent Transaction Analysis

A valuation method which compares the company being valued to similar companies from mergers and acquisitions transactions (relative valuation technique). The price paid for similar companies in the past, can be considered as an indicator of a company's value, and thus be used to estimate what a share of stock would be worth in the case of an acquisition. This type of analysis relies on publicly available information to create a reasonable estimate of multiples and premiums that others have paid for a publicly traded company. Understanding the relevance of a particular transaction is a vital component of precedent transaction analysis. Companies should be chosen based on having similar financial characteristics, industry, size, and type of transaction.

This type of analysis can be very difficult to conduct due to the lack of quality information relating to transactions. Unless you as an analyst are particularly lucky, in most cases market conditions, size of company and particular timing of transaction would be quite different and accounting for these discrepancies can be very difficult. Variables such as competitor size, brand recognition, market demand, business cycle, exchange rates, geopolitical effects, new legislation are just a few of the numerous factors which can have great effects on a particular valuation. Despite this there is some useful information which can be gathered. Firstly, precedent transaction analysis does provide a general assessment of the market's demand for a particular asset and approximate valuation of the asset. It can also be used in tandem with the valuation method discussed below to act as a basic idea of company value.

Comparable Company Analysis

Comparable company analysis (CCA), also referred to as comp analysis, derives metrics and ratios to compare similar companies to determine a fair value for the company in question. CCA works under the assumption that similar companies will have similar valuation multiples, such as P/E ratio. Conducting comp analysis is a fundamental skill for analysts and is fairly straightforward to implement.

Comparable company analysis starts with establishing a group consisting of companies of similar size, industry, and/or region. Investors are then able to compare a particular company to its competitors on a relative basis. This information can then be used to determine a company's enterprise value (EV) and to calculate other ratios used for comparison.

Key Company Statistics

In this section I will introduce some metrics and definitions which we will use in the valuation techniques outlined in this part of the course.

Market Capitalisation

Market capitalisation (market cap) is the total market value of a company's common and preferred shares outstanding. It is generally used as a figure to determine a company's size.

Market cap can be calculated using the equation below:

$$\text{Market Cap} = \text{Current Share Price} * \text{Total Number of Shares Outstanding}$$

Enterprise Value

Enterprise value (EV) is a measure of a company's total value and is often used as a more comprehensive alternative to market capitalisation. Enterprise value includes market capitalisation of a company but also short-term and long-term debt and any cash or cash equivalents on the company's balance sheet.

Components of EV

Market Cap – The total value of a company's outstanding common and preferred shares.

Debt – The sum of long-term and short-term debt.

Preferred Equity – Equity which has a higher priority of company's cashflow distribution.

Minority Interest – Stake less than 50% in a subsidiary that is controlled by a parent company.

Cash and Cash Equivalents – Total amount of cash and investments that can be readily converted into cash.

From these components we can calculate enterprise value as:

$$EV = \text{Market Cap} + \text{Total Debt} + \text{Minority Interest} + \text{Preferred Equity} - \text{Cash and Cash Equivalents}$$

Enterprise value is different from market cap in several ways and can be considered as a more accurate representation of a firm's value. EV tells investors how much another company would need if it wanted to purchase that company. Furthermore, in some cases EV can be negative if the total value of its cash and cash equivalents surpasses that of the combined total of its market cap and debts. This can be an indicator that the company is not using its assets very well but this depends heavily on the type of company and the industry.

Book Value

Book value is the value of the company according to the balance sheet. It is essentially the value of the company once all assets are liquidated, and liabilities paid off. The equation for book value is shown below:

$$\text{Book Value} = \text{Total Assets} - \text{Total Liabilities}$$

You can also calculate book value for common shares by simply taking away the value of preferred shareholder equity.

Liquidation Value

Liquidation value is a similar measurement to book value but it only takes into account physical assets. Liquidation value is the value of the physical assets of a company once they have been sold and all liabilities paid off. Therefore, we can calculate it as:

$$\text{Liquidation Value} = \text{Total Physical Assets} - \text{Total Liabilities}$$

Liquidation value tells the investor how much they are likely to receive if a company goes bankrupt and has to liquidate all its assets.

EBIT and EBITDA

EBIT refers to earnings before interest and tax and is an indicator of a company's profitability.

EBIT can be calculated using the formulas below:

$$EBIT = \text{Revenue} - \text{Expenses} + \text{Interest} + \text{Taxes}$$

$$EBIT = \text{Net Income} + \text{Interest} + \text{Taxes}$$

$$EBIT = \text{Revenue} - \text{COGS} - \text{Operating Expenses}$$

Where COGS refers to the cost of goods sold.

EBIT measures profit by ignoring taxes and interest expense, focussing solely on a company's ability to generate earnings from its operations by ignoring variables such as the tax burden and capital structure. EBIT comes in very handy when comparing two different companies as it allows us to focus solely on operating profit of the company without taxes being factored into the analysis. For example, if one company recent received a tax break or there was a cut in corporate taxes in a particular country, the company's net income or profit would increase. However, with EBIT all the benefits of the tax cut are removed from the analysis and the investor can fairly compare the two companies in the same industry but with different tax rates. The same is true when it comes to debt. A company in a capital-intensive industry will have a large amount of fixed assets on its balance sheet which are typically financed by taking on debt. For example, automobile firms are capital intensive because they need to spend a large amount of fixed assets such as machinery and factory real estate. As a result capital intensive industries have high-interest expenses due to a large amount of debt on their balance sheets. However, the debt, if managed property, is necessary for the long-term growth of companies. Companies in the same industry may have different amounts debt when compared to each other. As a result, companies would have differing interest expenses when compared to each other. This is where EBIT comes into play as it can help investors analyse companies based on operating performance and earnings potential while removing the effect of interest expenses and taxes.

EBIT can be taken a step further and a metric called EBITDA is introduced. This simply stands for earnings before interest, taxes, depreciation, and amortization. It takes EBIT and strip out the depreciation and amortization expense when calculating profitability.

EBITDA can be calculated using the following equation:

$$EBITDA = Revenue - Expenses + Interest + Taxes + Depreciation + Amortization$$

$$EBITDA = EBIT + Depreciation + Amortization$$

EBITDA is of significant value when looking at companies with significant amount of fixed assets as the depreciation expense can have a generate impact on net income. EBITDA measures a company's profits by removing depreciation and amortization further refining it as a measure for operational profitability.

Comparable Company Analysis Metrics

Earnings per Share

Earnings per share (EPS) of a company is equal to net profit minus preferred dividends divided by the number of common shares it has outstanding:

$$EPS = \frac{Net\ Income - Preferred\ Dividends}{Weighted\ Average\ Number\ of\ Common\ Shares\ Outstanding}$$

Where the weighted average number of common shares outstanding is the shares outstanding, each month weighted by the number of months those shares were outstanding. Simply put it is a more accurate measure for shares outstanding over a period of time.

Earnings per share is one of the most important metrics used when determining a firm's profitability as it indicates how much money a company makes for each share of its stock. EPS on its own isn't very useful but it is a major component of price-to-earnings ratio which we will cover below.

Price-to-Earnings (P/E)

Price to earnings (P/E) ratio is a ratio used for valuating a company by measuring its share price relative to earnings per share. This is more useful than earnings per share as it allows a like for like comparison between other companies since it reflects how much an investor would have to pay per unit of current earnings.

The formula for P/E ratio is:

$$Price\ to\ Earnings\ Ratio = \frac{Market\ Value\ per\ Share}{Earnings\ per\ Share}$$

P/E ratio helps determine whether a stock is overvalued or undervalued and can be benchmarked against other companies in the same industry or against the broader market too. Price-to-earnings ratio is one of the most commonly used metrics when it comes to valuations as it represents the price an investor needs to invest in order to receive \$1 of that company's earnings. For example, a P/E multiple of 25x indicates than an investor is willing to pay \$25 for \$1 of current earnings. It helps investors determine the market value of a stock as compared

to the company's earnings. A high price-to-earnings ratio could mean that the company's stock price is high relative to earnings and possibly overvalued.

Like most metrics, P/E ratio comes with some limitations and so must be used in tandem with other metrics to provide a better picture. Companies that aren't profitable, and consequently have no earnings, pose a challenge when it comes to calculating the P/E ratio and are often listed as 'N/A' on most data sources. Despite the company having zero or negative earnings, it may still have good future potential despite it not producing positive earnings at the moment. Furthermore, valuations and growth rates of companies often vary widely between sectors due to the different ways companies earn money and difference in timelines. Hence P/E ratio should only be used as a comparative tool when comparing companies in the same sector as it will produce a far better insight into the company in question.

EV/EBIT and EV/EBITDA

EV/EBIT or EV/EBITDA is a metric used as a valuation tool to compare the value of a company debt included, to the company's cash earnings less non-cash expenses. It is sometimes considered as a more accurate measure than using the price-to-earnings ratio.

$$EV/EBIT = \frac{\text{Enterprise Value}}{\text{Earnings before Interest and Tax}}$$

$$EV/EBITDA = \frac{\text{Enterprise Value}}{\text{Earnings before Interest, Tax, Depreciation, and Amortization}}$$

This metric can then be used to compare companies in a similar industry and gauge whether the company is over or under valued. Can only really use the ratio for companies in similar industries as the methods of generating earnings and thus their absolute value change depending on the company at hand.

Debt-to-Equity

If a certain company is borrowing too much this can increase fixed charges and reduce the earnings available for dividends and pose a risk for shareholders. The debt-to-equity ratio measures how a company is funding its operation using borrowed money. It is calculated using the following equation:

$$\text{Debt/Equity Ratio} = \frac{\text{Total Liabilities}}{\text{Total Shareholder Equity}}$$

Debt-to-equity ratio can indicate whether shareholder equity can cover total debt obligations and act as a metric to compare the amounts of leverage different companies have. Once again companies within the same industry should be used to compare this since the ratio varies between different industries. A company with a low debt to equity ratio can be seen as generally lower risk since they are less likely to default on loans however excessively low debt to equity ratios could indicate a company is not using leverage to its advantage.

Working Capital and Quick

Both the working capital ratio and quick ratio are measures of company liquidity. Liquidity refers to how easily a company can turn assets into cash to pay short-term obligations.

Working Capital Ratio

Working capital is the difference between a firm's current assets and current liability and represents a company's ability to pay its current liabilities with its current assets.

The working capital ratio is equal to a company's current assets divided by its current liabilities:

$$\text{Working Capital Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

If for example a company has a working capital ratio of 2, it can be an indication of healthy short-term liquidity. A working capital ratio of 1 can imply that a company may have liquidity problems, but this could only be a temporary issue depending on its future plans. Excessively high working capital ratio can mean that a company has too much in its short-term assets (e.g., cash) some of which could be better used to invest in the company or pay shareholder dividends.

Quick Ratio

This is also called the acid test and is another measure of liquidity. It represents a company's ability to pay current liabilities with assets that can be converted into cash quickly. The calculation for quick ratio is current assets minus inventory minus prepaid expenses divided by current liabilities:

$$\text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventory} - \text{Prepaid Expenses}}{\text{Current Liabilities}}$$

Formula removes inventory as it can take time to sell and convert inventory into liquid assets. Like the working capital ratio, a healthy company would have a ratio of around 1.5 indicating that the company has enough money to pay its bills and continue operating. A quick ratio less than 1 can indicate there isn't enough in liquid assets to pay short-term liabilities. The company may therefore have to raise capital.

Return on Equity

This is another metric for calculating profitability and measures how effectively a company uses shareholders' money to generate profit. Return on equity can be calculated using the following formula:

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Shareholders' Equity}}$$

A good return on equity is seen as one that increases over time and sometimes net income is used before the payment of dividends, as this would represent a return to investors.

Discounted Cash Flow Model

This brings us onto one of the most common absolute valuation models used by analysts which aims to work out the intrinsic value of a company. The discounted cashflow model is a valuation method used to estimate the value of an investment based on the present value of expected future cashflows.

Time Value of Money

The basis of a discounted cash flow model is the idea of *the time value of money*. The time value of money assumes that a £1,000 you have today is worth more than a £1,000 you receive in a year's time because it can be invested, and income can be produced from it.

The best way of understanding this is by taking two options, Option A is where you are set to receive £1,000 today and Option B is where you are set to receive £1,000 in 5 years' time. If we take Option B, then we have no potential for increasing the value of this £1,000. But if we take Option A, your future value would be the £1,000 plus any interest acquired over the five years. If we take the current market interest rate set at 4.5%, this will amount to £1,246.

This illustrates the time value of money and is a concept we will be using when it comes to bring values of future cashflows back to the present time.

Breakdown of a DCF Model

Free Cash Flow

There are two types of free cash flow: levered and unlevered. Both these types of cashflow appear on the balance sheet but have a slight difference. Unlevered free cash flow, also known as free cash flow to the firm (FCFF), is cash flow before any interest payments on debt obligations. Levered free cash flow also known as free cash flow to equity (FCFE) and is calculated after a company meets its debt obligations. The company can then use this money to pay dividends to shareholders, buy back stock, or invest in new business projects. A company that uses a lot of debt to finance its operations will have large amounts of unlevered free cash flow, therefore it must be compared to levered free cash flow to see how much free cash flow is being used to pay interest expenses. For a DCF, unlevered free cash flow is used as it can be very difficult to predict a company's future debt. From this point onward when free cash flow is mentioned assumed this is unlevered free cash flow.

Free cashflow is simply the cash available to a company to pay dividends or to repay creditors.

Free cashflow can be calculated by taking net income and subtracting investments made to maintain the company's assets and production:

$$\text{Free Cash Flow} = \text{Net Income} + \text{Non Cash Expenses} - \text{Change in NWC} - \text{Capital Expenditure}$$

$$\text{Free Cash Flow}$$

$$= \text{EBIT} * (1 - \text{tax rate}) + \text{Depreciation and Amortization} - \text{Change in NWC} - \text{Capital Expenditure}$$

Net Working Capital

Net working capital is the difference between the company's current short-term assets and its current short-term liabilities. This can be calculated by taking current assets from the balance sheet and subtracting current liabilities also on the balance sheet. However, to make this more accurate we will take some items from the cash flow statement too:

$$\text{Change in NWC} = \text{Change in Accounts Receivables} + \text{Change in Inventory} - \text{Change in Accounts Payable}$$

Capital Expenditure

Capital expenditure is funds used by the company to acquire, upgrade, and maintain physical assets. This can often be found on the company's cashflow statement but if it cannot be found we can also calculate it by the change in PP&E (Property, Plant, & Equipment) and subtracting depreciation during the current period

$$CapEx = PPE(\text{current year}) - PPE(\text{previous year}) - Depreciation$$

By using the equation outlined above we have free cash flow which is the first step of our discounted cash flow model. Free cash flow is then projected over the forecasting period using various assumptions we make as analysts. In this lecture we will not cover how exactly this free cash flow is projected, but we will assume that we have projected free cash flow over the forecast period. The exact methods on how to project free cash flow will be covered in the next lecture.

Terminal Value

The next step in our discounted cash flow model is to calculate the terminal value of the company at hand. The terminal value is simply the value of the company beyond the forecasted period when future cash flows can be estimated. Since you can't keep forecasting cash flows forever, at some point an assumption needs to be made about the cash flows beyond the final explicit forecast year by estimating a lump sum value of the business past this period. This lump sum is called the terminal value (TV) and can be calculated in 2 main ways.

Perpetuity Growth Rate

We assume that cash flows will grow at a stable rate forever starting at some point in the future (final year of projection). The terminal value is therefore calculated by dividing the last cash flow forecast by the difference between the discount rate and terminal growth rate:

$$TV = \frac{FCF_n \times (1 + g)}{d - g}$$

In this formula FCF_n represents the free cash flow for the last forecast period, g represents the perpetuity growth rate and d represents the discount rate. The perpetuity growth rate is a constant growth rate forecasted for the company after the final free cash flow period. In general, it is best to use the 10-year US government bond yield (also the Risk-Free Rate) and is currently at 3.6%. You can also take into account historical inflation (2-3%) and historical GDP growth rate (4-5%) to sense check the value you use. If you assume a perpetuity growth rate higher than 5%, you are saying that you expect the company's growth rate to outpace the economy's growth rate. Which, after a long period of time, ends up as the company being larger than the world economy! In regard to, d , the discount rate, we will cover what this means and how to calculate it below.

Exit Multiple

The exit multiple is another method in which you can calculate the terminal value of a company. This method assumes that the business will be valued at the end of the projection period based on public market valuations. It is typically calculated by applying an appropriate multiple (generally EV/EBITDA, EV/EBIT etc) to the relevant statistic projected for the last projected year. This specific multiple is generally found using comparable company analysis and multiplied by the projected statistic to arrive at your enterprise value at the end of the forecasting. The equation to calculate terminal value using an exit multiple is shown below:

$$TV = \text{Last 12 Multiple} \times \text{Exit Multiple}$$

A more specific equation for TV using EBITDA may look like

$$TV = \text{Final Year EBITDA} \times \text{EV/EBITDA Multiple}$$

The exit multiple does use relative valuation in an inherently absolute valuation technique but for industries well covered by analysts it can yield a more reliable terminal value when compared to the perpetuity growth method. The assumptions when using an exit multiple can generally be easier to defend and better explained but this is reliant on the fact that data for the exit multiple is available.

Discount Rate

So now that we have our projected free cash flow and our terminal value at the end of the forecasting period, we now need to come up with a rate to discount all these future cash flows to the present. The discount rate represents the expected return on an investment based on its risk profile (meaning that the discount rate is a function of how risky the cash flow is). It can also be considered as the minimum threshold return of an investment based on comparable investments with similar risks. A higher discount rate makes a company's cashflows less valuable, as it implies the investment carries a greater amount of risk. In order to calculate this rate, we used something called the weighted average cost of capital, or WACC for short.

WACC – Weighted Average Cost of Capital

The WACC measures how much the company pays to gain capital to invest into the company and is hence often used as the discount rate. It can be viewed as the opportunity cost of an investment based on comparable investments with similar risk profiles. It is calculated by multiplying the equity weight by the cost of equity and add it to the debt weight multiplied by the tax affected cost of debt. This will make a bit more sense when we look at the formula.

$$WACC = \left(\frac{\text{Equity}}{\text{Debt} + \text{Equity}} \times R_{\text{equity}} \right) + \left(\frac{\text{Debt}}{\text{Debt} + \text{Equity}} \times R_{\text{debt}}(1 - t) \right)$$

Where:

$\frac{\text{Equity}}{\text{Debt} + \text{Equity}}$ and $\frac{\text{Debt}}{\text{Debt} + \text{Equity}}$ = the weights of debt and equity in the company's total capital structure.

R_{equity} = cost of equity

R_{debt} = is the cost of debt, and must be tax-affected since interest is tax deductible

t = Tax rate

Each of the components in the WACC equation must also be calculated.

Cost of Equity

The cost of equity is the expected return for equity investors in exchange for the risk they feel they are taking when investing. We use the capital asset pricing model to work out the cost of equity and can be written as follows:

$$\text{Cost of Equity} = \text{Risk Free Rate} + (\beta \times \text{Equity Risk Premium})$$

The risk-free rate is the theoretical rate of return of an investment with zero risk. For US companies we just use the 10-year-US government bond yield (3.63% now).

The β (beta) measures the systematic risk of a company compared to the market. It measures a company's sensitivity to market volatility. It is equal to the covariance between expected returns on an asset and the market,

divided by the variance of expected returns on the market. This course won't go into how specifically it is calculated but values for beta can generally be found in most financial data sites. A company with a beta of 1.0 would expect market returns consistent with the rest of the market. For example, if the market were to go up by 10%, the company should see a return equal to 10%. Whereas a company with a beta of 2.0 would see an increase of return on 20%.

Beta/ Market Sensitivity Relationship

$\beta = 0$ – No Market Sensitivity

$\beta = 1$ – Equal Market Sensitivity

$\beta < 1$ – Low Market Sensitivity

$\beta > 1$ – High Market Sensitivity

$\beta < 1$ – Negative Market Sensitivity

Equity Risk Premium (ERP) is the premium paid when you invest in the stock market vs risk-free products such as government bonds. This accounts for the difference between the riskier stock market option vs a less risky product such as government bonds.

$$\text{Equity Risk Premium} = \text{Expected Market Return} - \text{Risk Free Rate}$$

Cost of Debt

Cost of debt is far easier to find than the cost of equity as it is the rate of return lenders expect to be compensated for the potential loss of capital. It can be effectively thought of as the interest rate the company pays on its long-term debt obligations. It can be calculated using the following formula:

$$\text{Cost of Debt} = \frac{\text{Annual Interest Expense}}{\text{Total Debt}} \times (1 - \text{Tax Rate})$$

We multiply the fraction by the (1- Tax Rate) as interest expenses are tax deductible so the effect cost of debt needs to remove the tax-deductible value of interest. In order to find this tax rate, we can look at the income statement as it typically appears on the line above net income.

Finally, we have all the values necessary to discount all the cash flows back to the present day value.

Enterprise Value and Equity Value

We discount the cash flows using the following formula:

$$\text{Enterprise Value} = \frac{FCF_1}{(1+r)^1} + \frac{FCF_2}{(1+r)^2} + \dots + \frac{FCF_n}{(1+r)^n} + \frac{TV}{(1+r)^n}$$

Where TV can be calculated using your perpetuity growth model or exit multiple and r refers to the discount rate. Now that we have enterprise value, we need to convert it to equity value before finding our per share price. Equity value is how much of the company value belongs to equity holders. This is done by subtracting net debt and other non-equity claims and adding back the value of non-operating assets such as cash or investments.

$$\text{Equity Value} = \text{Enterprise Value} - \text{Net Debt}$$

Where Net Debt is comprised of:

$$\text{Net Debt} = \text{Total Non Equity Claims} - \text{Total Non Operating Assets}$$

Price per Share

Finally to work out the price per share, what we were after all this time, we divide our equity value by the number of diluted shares outstanding as of the valuation date. Then we simply compare the current price with our intrinsic value.

$$\text{Calculated Price per Share} = \frac{\text{Equity Value}}{\text{Number of Shares Outstanding}}$$

There we have it! The value of projected future cash flows today.

Sensitivity Analysis

As you have seen there a vast number of assumptions we have made to arrive at the final price of the share. In order to conduct a thorough analysis, investors also undertake a sensitivity analysis on the assumptions made. By adjusting these number we can create sensitivity table which allows for a better data set when making the decision of whether or not to invest in a company.